OmniCast

System Initialization: Enterprise Media, Real-Time LiveStream
& Broadcast Intelligence Network (Fall 2026)



📡 PRN232 Backend & Web Portal

[STATUS: ONLINE]
[LATENCY: <5ms]
[DATABASE: OK]



📱 PRM393 Mobile Edge Application

[BUILD: V2.4.1-RC]
[DEVICE: EDGE]
[SYNC: ACTIVE]



🧠 AI Broadcast Intelligence

[MODEL: GPT-4V-BROADCAST]
[PROCESSING: REAL-TIME]
[ANALYTICS: ENGAGED]



⚡ High-Fidelity Live Studio

[STREAM: 4K HDR 60FPS]
[AUDIO: ATMOS]
[BANDWIDTH: STABLE]

Developed at FPTU Quy Nhon AI Campus — FU Cahata. Supervised by Mentor Thầy Huỳnh Vang.

Gemini Notebook

The Engineering Taskforce & Domain Ownership

PROFILE: TOMOUTFIT



Nguyễn Bình An (TomOutfit)

Team Leader & Core Backend

- NestJS 10
- RBAC Security
- EPG Engine

PROFILE: QE170080



Trần Thanh Sơn (QE170080)

Microservices & Database

- gRPC/HTTP2
- Supabase PostgreSQL
- Telemetry

PROFILE: QE180136



Nguyễn Vĩnh Phú (QE180136)

Mobile Flutter Lead

- Flutter 3.x
- Offline SQLite
- Mobile Live

PROFILE: TENG122



Đồng Hoàng Nguyên (teng122)

Web, Live & AI

- Next.js 15
- GPT-4o AI
- Socket.IO

Gemini Notebook

The Industry Collision Course: Fragmentation & Dead Air

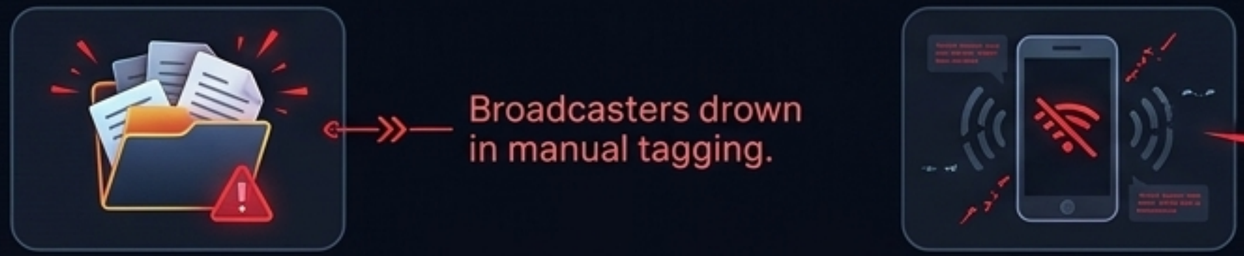
Top Pipeline:
Multi-Source
Live Complexity ⚠



Middle Pipeline:
Airtime Collisions
& 24/7 Dead Air



Manual Curation
& Rigid Mobile



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Unified Hybrid Ecosystem: The OmniCast Matrix



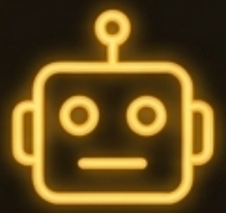
4-Mode Live Studio

Curated VOD, WebRTC WHIP Camera, and Public Catalog with zero-downtime program switching via Socket.IO /live.



Mathematical Collision Engine

Single raw SQL interval query paired with a deterministic, seeded 24/7 EPG auto-filler.



AI Broadcast Curator

Single-call GPT-4o curation utilizing Zod schema validation backed by a localized heuristic fallback system.

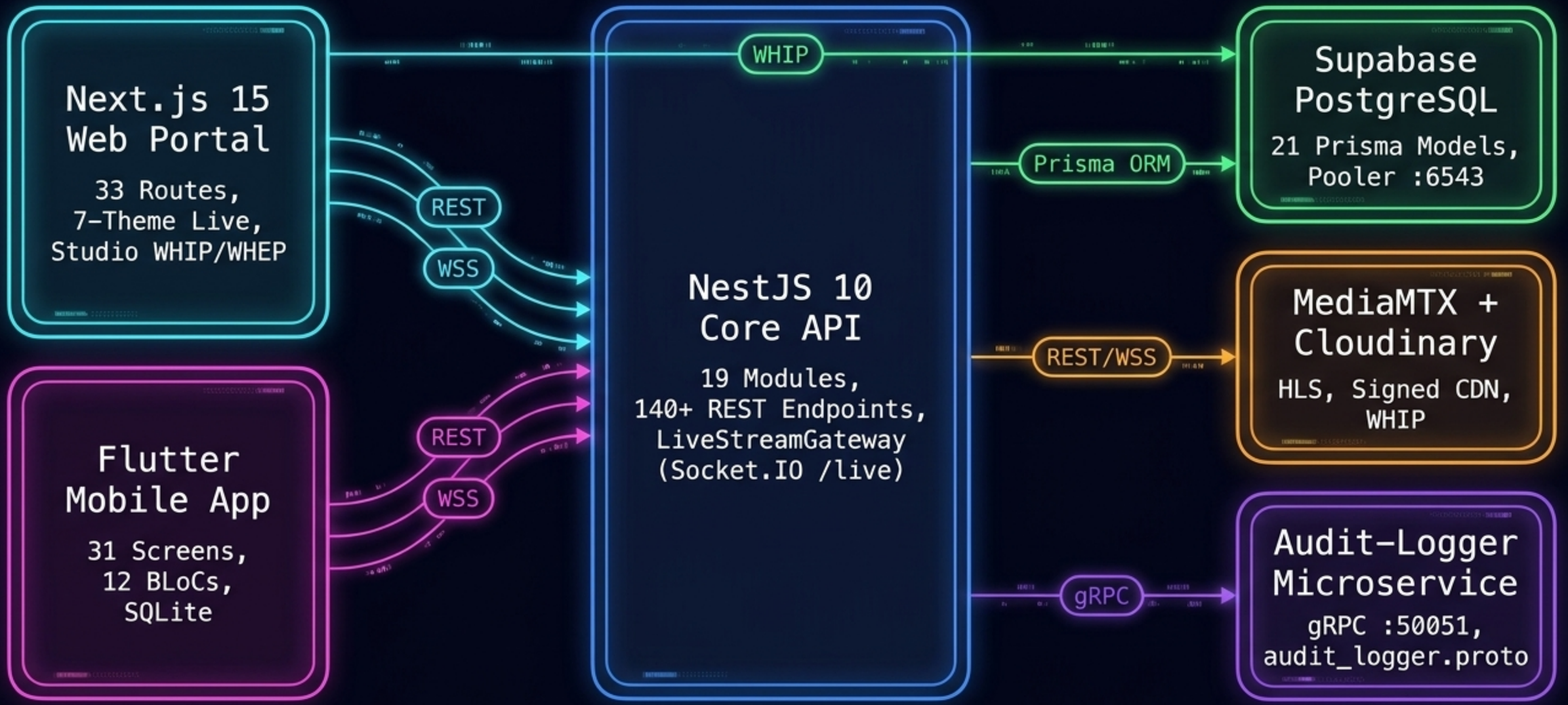


LAN-Aware Offline Mobile

SQLite mutation queue with automatic Wi-Fi MediaMTX LAN discovery for seamless offline-first resilience.

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End-to-End Dual-Platform Unified Topology



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PRN232 Backend Brain & RBAC Security Vault

19 Modules & 5-Domain Database



Powered by NestJS 10 & 21 Prisma Models

Security Vault Matrix

- ADMIN: System governance, gRPC audit log explorer.
- STAFF: Live Studio broadcasting, AI curation, scheduling.
- VIEWER: Live room chat, 6-tier Super Chater Chat, watchlists.
- GUEST: Public EPG grid & VOD discovery.

Defense in Depth via Helmet, CORS whitelist, Bcrypt (12 rounds), and JWT (60m + 90d rotation).

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Algorithmic Rigor: Collision Prevention & 24/7 EPG Engine

THE MATHEMATICAL TIMELINE



CODE SNIPPET: SQL COLLISION CHECK

```sql
SELECT id, title FROM "LiveEvent"
WHERE "channelId" = $1::uuid
  AND status IN ('SCHEDULED', 'LIVE')
  AND "scheduledAt" < $newEnd
  AND COALESCE("endedAt", "scheduledAt" +
    ("duration" || 'min')::interval) > $newStart;
```

SEEDED 24/7 ENGINE: TECHNICAL READOUT

- **Zero Empty Days:** Guarantees full 24-hour coverage.
- **Seeded PRNG:** Keyed on (date, channelId) ensuring deterministic cross-client playout.
- **Pure Function:** Zero I/O side effects, verified by 18 unit tests.

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Broadcast Intelligence: AI Curator & gRPC Audit Telemetry

AI Validation Pipeline

Program Input
(Title/Tags)

OpenAI GPT-4o
(Prompt v4.0.0, temp=0.3, top_p=0.95, cuts latency 66%)

Zod Schema Check
(Strict Type Validation)

VALIDATED OUTPUT

Parse Fail

Local Rule-Based Heuristic
(Sentiment / Compliance)

gRPC Audit Microservice

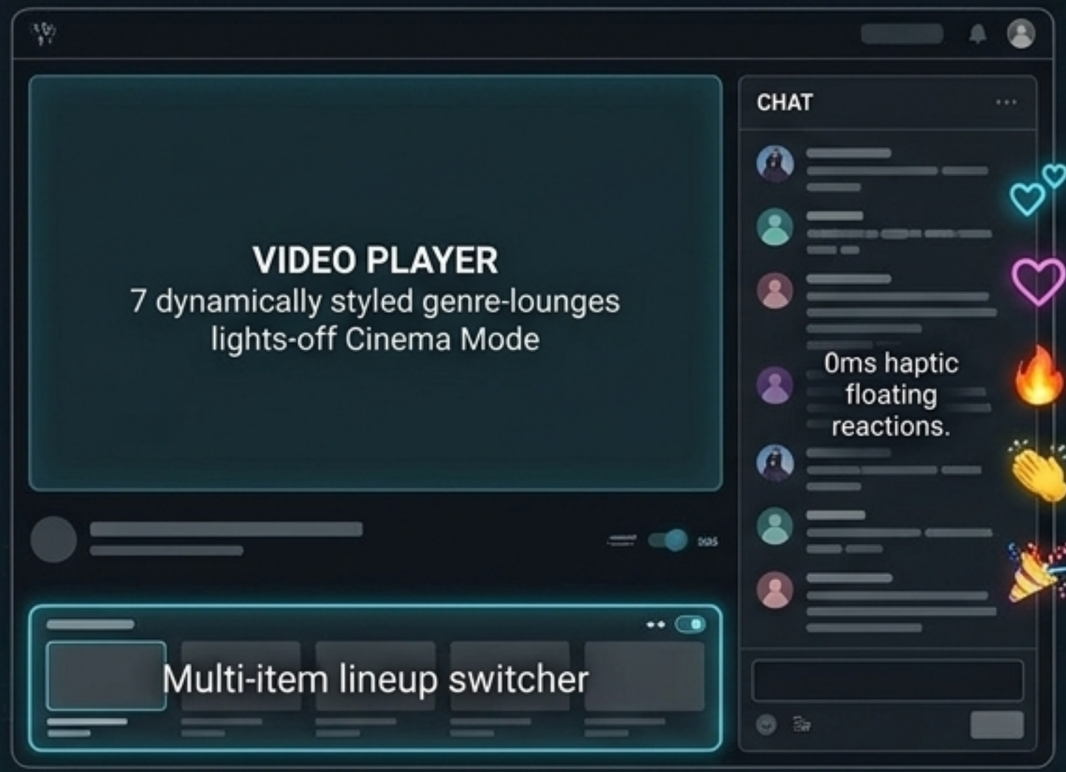


- Standalone HTTP/2 telemetry (Port 50051).
- Exposes LogAudit over binary Protocol Buffers.
- Non-blocking fire-and-forget architecture keeps main API zero-latency.

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Next.js Command Center: Web Portal & Live Studio

Interactive Live Room (/live/[id])



4-Tab Broadcast Console (/studio/live)



Stack: Next.js 15 App Router (33 Routes) + Socket.IO /live

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PRM393 Mobile Edge: Flutter Clean Architecture

PRESENTATION (31 Screens & 22 Custom Widgets)

6-tab `MainNavigationShell` and `GoRouter` topology.

LOGIC (12 BLoC/Cubits)

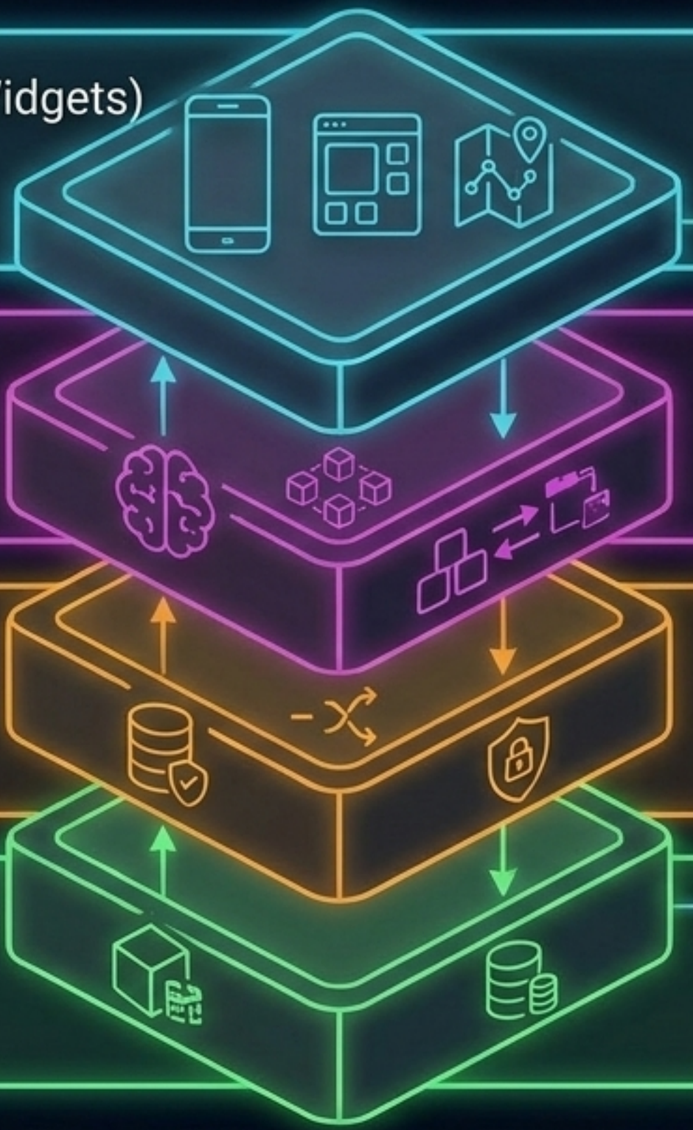
`Auth`, `Channel`, `Epg`, `LiveEvents` featuring droppable/restartable event transformers.

REPOSITORIES (10 Repositories)

Custom `Dio` `AuthInterceptor` with 401 queue locking.

LOCAL STORAGE

SQLite `omnicast_local.db`



Built on Flutter 3.x and Dart 3.

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OmniPlayer Networking: LAN Discovery & Offline Resilience

Connectivity Flowchart

Network UP

Automatic Wi-Fi LAN Discovery

`LanDiscoveryService` scans the local /24 subnet on `port 8888`. Auto-replaces localhost with the real MediaMTX host IP on physical devices.

Zero-Cartoon Live Retry

executes 10x exponential backoff.

Network DOWN

Offline-First SQLite Cache

Reads seamlessly from local DB.

Pending Mutation Queue (`pending_ops`)

Captures actions and replays them FIFO upon internet reconnection.

Bonus Feature: 15-Minute Advance local OS alarms for bookmarked shows

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The Mobile Field Reporter Viewfinder HUD



SMPTE Timecode (HH:MM:SS:FF) real-time display.

Flashing Red ON-AIR Tally Border.

Rule-of-Thirds & Safe Area composition grid.

Auto-scrolling Reporter Teleprompter.

Concept-Telemetry Audio Composition.

Dual-channel Stereo VU Audio Meter.

Dual Broadcast Modes

- **Push-to-Live:** Real-time WebRTC WHIP
- **Record-then-Live:** 1080p MP4 + Master QC Review

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System Diagnostics: Cross-Platform Synergy Matrix

Dimension	PRN232 (Web/API)	PRM393 (Mobile)
Core Stack	NestJS 10 / Next.js 15	Flutter 3.x / Dart 3
Architecture	19 Domain Modules + gRPC	4-Tier Clean Architecture
LiveStream	7-Theme Live Room + Web WHIP Studio	LiveWatchScreen + ENG Recorder
Code Volume	79,568 Physical LOC	53,922 Physical LOC
Automated Tests	391 Tests (163 Backend, 228 Web)	91 Tests (56 Unit, 35 Widget)
Pass Rate	✓ 100% PASS (+ 8 E2E)	✓ 100% PASS

TOTAL SYSTEM SCALE: 142,897 Lines of Code • 482/482 Tests Pass

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Live Production Telemetry & System Deployment

Web Portal: Vercel Live Environment

omnicast-fe.vercel.app

Core API: Swagger Telemetry Documentation

omnicast-api.vercel.app/swagger

Mobile Client: APK Release v1.2.0

(GitHub Releases)

Source Control: Official Monorepo

(GitHub)

Credentials Bar: Pre-configured Evaluation Accounts. Password: Admin123! ➡ admin@omnicast.tv (Full RBAC) ➡ staff@omnicast.tv (Live Studio) ➡ viewer1@omnicast.tv (Live Chat & Super Chat)

OmniCast is fully deployed and ready for Live Demonstration. FPTU Quy Nhon AI Campus — FU Cahata.

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